



AI & IOT INTEGRATED WIND TREE FOR OPTIMIZED MICRO WIND ENERGY

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ABSTRACT

- This project explores an innovative, nature-inspired approach to renewable energy generation by developing a wind tree system utilizing aero leaves.
- Designed to mimic the structure and aesthetics of a natural tree, this setup integrates small, lightweight aero-leaf turbines in place of leaves to capture wind from all directions.
- With the integration of “Internet of Things technology and AI-based monitoring”, the system ensures efficient energy tracking, fault detection, and real-time performance optimization.
- This model emphasizes affordability and accessibility, promoting clean energy adoption at the grassroots level.



INTRODUCTION

- Wind Tree technology is a nature-inspired solution that mimics the form and function of natural trees to harness wind energy efficiently using compact vertical axis turbines called aeroleaves.
- Each one acts as a miniature wind turbine that can generate electricity even from lowspeed winds in all directions, making it suitable for areas where traditional turbines are inefficient.
- Integrating Artificial Intelligence (AI) and Internet of Things (IoT) enhances real-time monitoring, predictive maintenance, and adaptive energy optimization in the system.
- This Project explores how combining smart systems and renewable energy principles can lead to a scalable and eco-friendly power generation solution.



LITERATURE SURVEY

REF. NO	TITLE	TECHNIQUE USED	PROS	CONS
1	Fabrication and performance analysis of the aero-leaf savonius wind turbine tree	Savonius rotor design, performance testing	Works efficiently, urban-friendly	Mechanical imbalance, require external battery
2	Nature-inspired designs in wind energy	Bio-mimicry analysis, computational simulation	Low-noise, useful for urban architecture	May increase fabrication complexity and cost
3	Small wind turbines and their potential for internet of things applications	Design & Evaluation of Small Wind Turbines	Sustainable Power Source for IOT	Wind Availability Constraints Low Power Output
4	Energy prediction of wind turbine using iot	Sensor Integration with IoT	Real-time Monitoring, Energy Forecasting	Limited Sensor Accuracy, Model Limitations
5	Optimizing Energy Flow in a Wind-PV-Battery Microgrid Using AI Techniques	Microgrid, Energy Management, Swarm Intelligence, Wind Power, Photovoltaics	Optimized energy flow means reduced dependency on expensive fossil-fuel backups and lower operational costs	Installing wind turbines, solar panels, battery systems, and sensors, along with AI infrastructure, requires a significant upfront investment.



LITERATURE SURVEY

REF. NO	TITLE & AUTHOR	TECHNIQUE USED	PROS	CONS
6	A comprehensive IoT cloud-based wind station ready for real-time measurements and artificial intelligence integration	Weather station Wind measurements IoT wind station Cloud-base weather monitoring	Immediate awareness of wind conditions and turbine performance	Requires investment in IoT hardware, cloud services, and AI development.
7	Review of Artificial Intelligence-Based Design Optimization of Wind Power Systems	Evolutionary Algorithms (GA, PSO), Hybrid AI Approaches	Comprehensive coverage of optimization areas	Limited discussion on real-time deployment
8	AI-Enabled Metaheuristic Optimization for Predictive Management of Renewable Energy Production in Smart Grids	Hybrid Metaheuristics (LSTM-RL, RL-SA, CNN-PSO)	Suitable for load balancing and rural microgrids	May require significant training data
9	Internet of Things (IOT) Based Machine Learning Techniques for Wind Energy Harvesting	IoT Sensors, Wireless Sensor Networks (WSNs)	Predictive Maintenance, Energy Output Optimization	High Initial Cost, Data Security Risks
10	Small Wind Electric System Energy Saver	Permanent Magnet Generators (PMG)	Energy Independence, Easy Installation	Noise & Vibration, Limited Output



MOTIVATION

- Most wind turbines require large open areas and consistent wind flow, making them unsuitable for cities, rooftops, or rural households with limited land.
- Inspired by natural leaf structures, the concept of a "Wind Tree" using aero-leaves offers a new path that is both functional and aesthetic.
- There is limited practical implementation data on nature-inspired vertical axis turbines like the aero-leaf design making this an opportunity to contribute real-world testing and improvement.
- The project aligns with global efforts to promote green, decentralized, and noise-free energy solutions that are safe for humans, animals, and the environment.



OBJECTIVE

- The Main motive of this project is to design and fabricate a low-cost Wind Tree prototype that harnesses wind energy using multiple aero-leaves arranged in a tree-like structure.
- To integrate IoT sensors for real-time monitoring of wind speed, turbine RPM, and energy generation statistics.
- To apply basic AI or Machine Learning models to predict energy output patterns and optimize blade performance under varying wind conditions.



EXISTING METHOD

- Earlier introduced project include a wind turbine model shaped like a tree, using aéroleaf shaped Savonius rotors as the main energy-generating components.
- Each aéro-leaf works as a small vertical-axis wind turbine (VAWT), capturing wind from all directions.
- The design focuses on low wind speed environments, making it ideal for urban and semi-urban use.
- The rotors are connected to small DC generators to produce electricity.

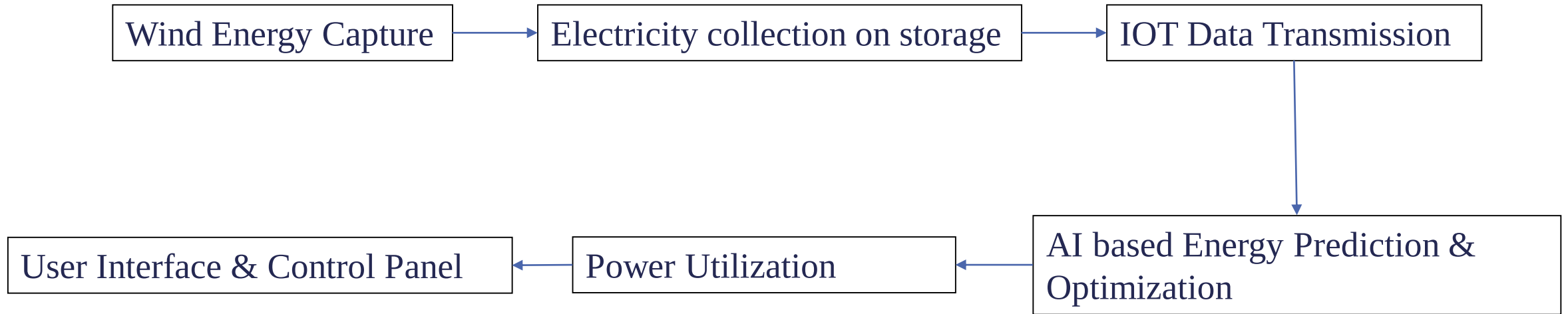


PROPOSED METHOD

- Design and build a small-scale Wind Tree using aero-leaf-shaped Savonius rotors.
- Using recycled plastic, DC motors to keep minimal or zero cost.
- Integrating IoT sensors to monitor wind speed, power output, and rotor performance in real time.
- Develop a machine learning model to predict energy generation based on historical weather and turbine data.



BLOCK DIAGRAM





EXPLANATION OF PROPOSED WORK

What is the use?

- ✓ This project focuses on designing a nature-inspired wind energy harvesting system called a Wind Tree, helps reduce dependence on fossil fuels by using naturally available wind.
- ✓ Unlike traditional wind turbines, this system is compact, silent, and suitable for urban and rural areas where space and aesthetics are concerns.
- ✓ The system integrates Artificial Intelligence (AI) for efficient wind prediction and system optimization, while IoT (Internet of Things) is used for real-time monitoring of power generation, turbine health, and environmental data.



EXPLANATION OF PROPOSED WORK

How it is Going to be used in your Project?

- ✓ Assemble a small wind tree with aeroleaves, ESP32, sensors (wind, power, vibration, temp), and actuators for control.
- ✓ Connect nodes via Wi-Fi/LoRa/NB-IoT; send data using MQTT/HTTP; create a live monitoring dashboard.
- ✓ Gather wind and power data, then train ML models for forecasting, power prediction, anomaly detection, and control optimization.
- ✓ Run lightweight AI on the microcontroller for real-time decisions, while heavy training/analytics run in the cloud.
- ✓ Apply closed-loop AI control, measure improvements in energy output and reliability, and refine the system through iterations.



EXPLANATION OF PROPOSED WORK

Why You Choose and explain the advantages ?

- ✓ With rising pollution and climate change, there's a strong need to shift from fossil fuels to ecofriendly energy solutions like wind power.
- ✓ Unlike big wind turbines, this project works even with low or irregular wind flow, which is ideal for cities and villages.
- ✓ The idea of using a "wind tree" with aero-leaves is unique, visually appealing, and blends well with the environment.
- ✓ This project can be built using low-cost, reused items, making it affordable for students and communities.

HARDWARE & SOFTWARE INVOLVED



HARDWARE:

- DC Micro Generators
- Battery
- Microcontroller
- Charge Controller
- Anemometer
- RPM Sensor
- Voltage & Current Sensors
- Wi-Fi/Bluetooth Module
- Mounting Tree Structure

SOFTWARE:

- ✓ Arduino IDE

For programming microcontrollers and reading sensor data.

- ✓ ThingSpeak

Cloud-based platform for real-time IoT data visualization.

- ✓ Proteus

For circuit simulation before real hardware testing.

- ✓ Python / MATLAB

Used for data analysis, predictions, and AI-based forecasting.

- ✓ MicroPython

For coding the microcontroller logic.



APPLICATION & ADVANTAGES

APPLICATION:

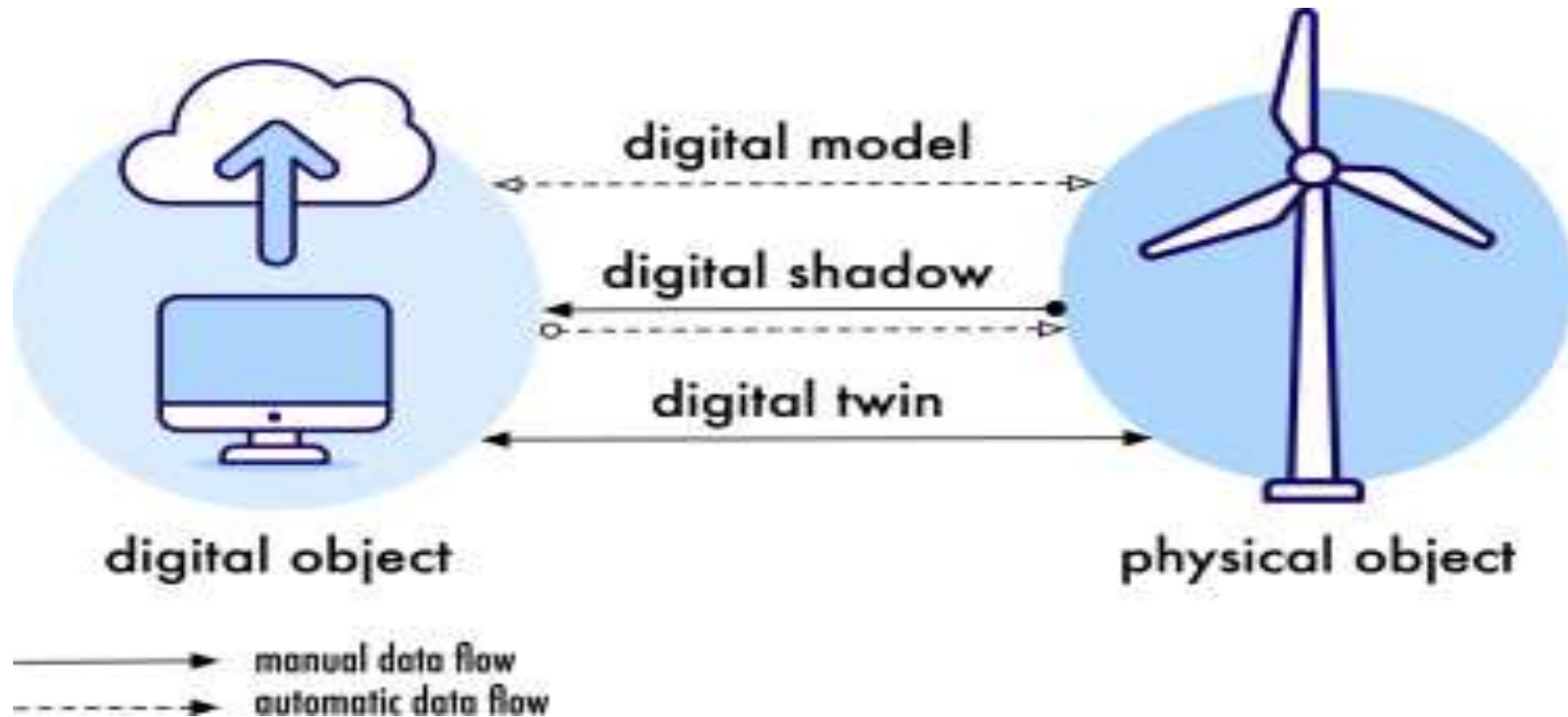
- This project supports sustainable infrastructure by powering streetlights, CCTV cameras, and IoT devices.
- It can provide basic electricity in remote villages or forest surveillance posts.
- It can acts as a natural energy backup for LED bulbs, fans, mobile charging, etc.
- It can be used in Environmental Monitoring System.



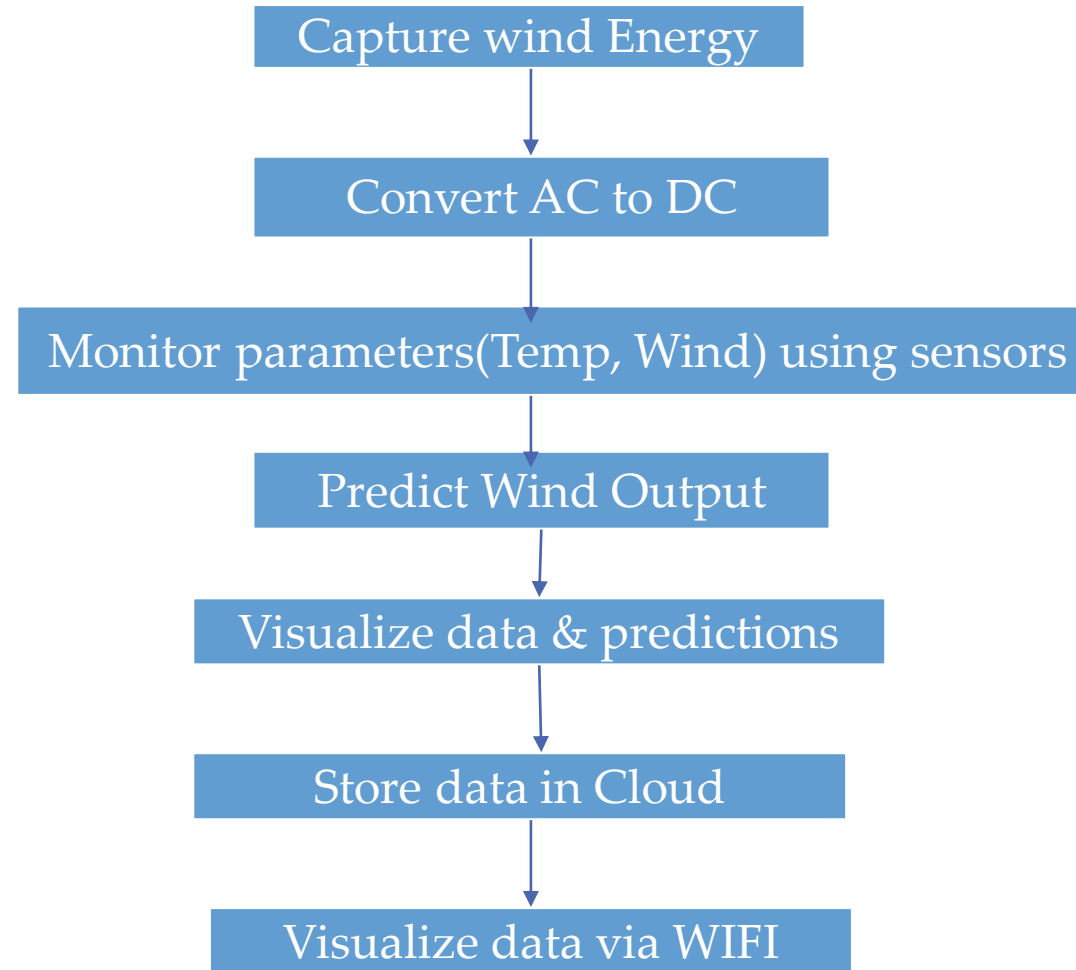
ADVANTAGES

- It works silently unlike traditional horizontal turbines.
- We can increase the number of leaves (turbines) easily as per energy needs.
- Real-time power prediction, fault detection, and performance tracking from your phone or laptop.
- It blends well in public places without disturbing the visual appeal.

SYSTEM ARCHITECTURE



UML DIAGRAM



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THANK YOU